

sintaxismark: complete user documentation

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1 Purpose and scope

`sintaxismark` is a LaTeX package for marking syntactic constituents, grammatical functions, heads, and annotated spans directly over running text. It is designed for teaching materials, exercises, handouts, slides, and compact linguistic analyses where a full syntactic tree would be too heavy.

The experimental monolithic version consists of a single file, `sintaxismark.sty`. This file contains the classic TikZ-overlay implementation, manual multiline helpers, experimental automatic splitting commands, and the experimental LuaLaTeX grid renderer.

2 Compilation and installation

Copy `sintaxismark.sty` to the same directory as your document, or install it in a local TeX tree.

```
\documentclass{article}
\usepackage{sintaxismark}
\begin{document}
Text.
\end{document}
```

If you use `gb4e`, load `gb4e` first and disable automatic math mode:

```
\documentclass{article}
\usepackage{gb4e}
\nomath
\usepackage{sintaxismark}
\begin{document}
\begin{exe}
\ex \SES{Juan} \PVS{\NV{compró} \OD{manzanas}}.
\end{exe}
\end{document}
```

Classic mode uses TikZ with `remember picture` and `overlay`. Compile twice if the first run places labels incorrectly. Grid mode and the automatic splitting commands require LuaLaTeX.

3 Modes of operation

1. **Classic mode:** commands such as `\spanline`, `\openbox`, `\tagbelow`, `\synul`, and the shortcut commands draw annotations directly around the text. This is the most flexible mode for short examples and fine manual control.
2. **Manual multiline mode:** helpers such as `\spanfirst`, `\spanmid`, `\spanlast`, `\boxfirst`, `\boxmid`, and `\boxlast` split a long annotation into visually separate segments.
3. **Automatic split mode:** `\autospanline` and `\autoopenbox` split plain text automatically. This is experimental and only parses plain text.
4. **Grid mode:** the `sintaxisgrid` environment parses nested `sintaxismark` markup, computes line breaks, and renders the whole example as one grid. It is intended to avoid diagonal lines in long multiline analyses.

4 Visual settings inherited from the classic implementation

The original spacing of `\spanline` and `\openbox` is controlled by the `/sintaxis` key family. The following defaults are the reference values used by the classic implementation.

Key	Default	Meaning
<code>color</code>	black	Annotation color.
<code>linewidth</code>	0.6pt	Stroke width for spans, boxes, and underlines.

<code>liney</code>	<code>1ex</code>	Vertical length of the legs of upper spans.
<code>linegap</code>	<code>5.5pt</code>	Gap between the text and the lower end of an upper span leg. The upper span line is placed at <code>linegap + liney</code> .
<code>labsep</code>	<code>1ex</code>	Gap between an upper span line and its label.
<code>boxd</code>	<code>3ex</code>	Depth of lower open boxes.
<code>boxsep</code>	<code>1.0ex</code>	Gap between a lower open box and its label.
<code>labelsep</code>	<code>unset</code>	General label separation override. If set, it overrides <code>boxsep</code> and <code>labelsepboxes</code> .
<code>labelsepboxes</code>	<code>unset</code>	Label separation override for lower boxes.
<code>nestedgap</code>	<code>0.3ex</code>	Extra gap used when nested lower annotations require additional depth.
<code>labheight</code>	<code>1.1ex</code>	Estimated label height used in vertical reservation.
<code>tagy</code>	<code>1.8ex</code>	Vertical distance for floating lower tags.
<code>tagx</code>	<code>0pt</code>	Horizontal displacement for floating lower tags.
<code>underw</code>	<code>0.8pt</code>	Underline stroke width.
<code>autogap</code>	<code>0.6ex</code>	Extra gap used by automatic depth calculation for nested lower boxes.
<code>leftend</code>	<code>true</code>	Draw the left vertical end of a span or open box.
<code>rightend</code>	<code>true</code>	Draw the right vertical end of a span or open box.
<code>autodepth</code>	<code>true</code>	Let nested lower annotations increase the depth of an outer open box.
<code>leftarrow</code>	<code>false</code>	Add a left arrow tip to the horizontal line.
<code>rightarrow</code>	<code>false</code>	Add a right arrow tip to the horizontal line.
<code>arrowlen</code>	<code>1.3ex</code>	Arrowhead length.
<code>arrowtip</code>	<code>Latex</code>	TikZ arrowhead type.
<code>arrowscale</code>	<code>1</code>	Arrowhead scale.
<code>arrowinset</code>	<code>0pt</code>	Arrowhead inset.
<code>labelalign</code>	<code>center</code>	Label alignment for lower boxes: <code>left</code> , <code>center</code> , or <code>right</code> .

The compact reservation layer currently reserves a fixed amount of vertical space: `4.5ex` below the line and `7.5ex` above the line. This is a pragmatic setting that prevents frequent collisions in examples embedded in `gb4e`.

Example. Changing classic spacing globally inside a document

```
\setsintaxischema{docencia}
```

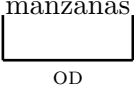
Example. Changing classic spacing locally for one annotation

```
\spanline[liney=3ex,linegap=0.8ex,labsep=1.2ex,linewidth=1pt]{pvs}{compró manzanas}
\openbox[boxd=3.6ex,boxsep=1.4ex,linewidth=1pt]{od}{manzanas}
\tagbelow[tagy=2.2ex,tagx=0.5ex]{md}{las}
```

(1)

PVS

compró manzanas

- (2) 
- (3) 

5 Visual presets

The package option selects a visual preset for classic mode:

```
\usepackage[docencia]{sintaxismark}
\usepackage[impresion]{sintaxismark}
\usepackage[diapositivas]{sintaxismark}
\usepackage[publicacion]{sintaxismark}
```

Presets may also be selected inside the document:

```
\setsintaxisschema{diapositivas}
```

Preset	Settings
docencia	linew=0.9pt, liney=2.4ex, linegap=0.6ex, labsep=1.0ex, boxd=3.2ex, boxsep=1.0ex, tagy=1.8ex, underw=1.0pt.
impresion	linew=0.5pt, liney=1.8ex, linegap=0pt, labsep=0.6ex, boxd=2.6ex, boxsep=0.8ex, tagy=1.6ex, underw=0.7pt.
diapositivas	linew=1.2pt, liney=3.0ex, linegap=0.8ex, labsep=1.2ex, boxd=3.6ex, boxsep=1.4ex, tagy=2.0ex, underw=1.2pt, arrowlen=1.8ex, arrowscale=1.25, labelsepboxes=1.6ex, nestedgap=0.8ex, labheight=1.2ex.
publicacion	linew=0.48pt, liney=2.0ex, linegap=0.3ex, labsep=0.7ex, boxd=2.8ex, boxsep=0.85ex, tagy=1.65ex, underw=0.65pt, arrowlen=1.2ex, arrowscale=1.0, labelsepboxes=0.9ex, nestedgap=0.45ex, labheight=1.0ex.

6 Core classic commands

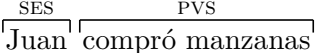
6.1 Upper spans: `\spanline`

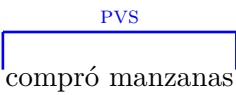
```
\spanline[<keys>]{<label>}{<text>}
```

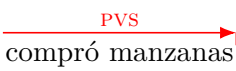
`\spanline` draws an upper horizontal line with optional vertical ends and a label above it.

Example. Upper spans

```
\spanline{ses}{Juan} \spanline{pvs}{compró manzanas}
\spanline[color=blue,linew=1pt,liney=3ex,labsep=1.2ex]{pvs}{compró manzanas}
\spanline[leftend=false,rightarrow,color=red]{pvs}{compró manzanas}
```

- (4) 

(5) 

(6) 

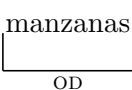
6.2 Lower open boxes: `\openbox`

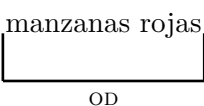
```
\openbox[<keys>]{<label>}{<text>}
```

`\openbox` draws a lower open bracket-like box with a label below it.

Example. Lower open boxes

```
\openbox{od}{manzanas}
\openbox[boxd=3.8ex,boxsep=1.4ex,linewidth=1pt]{od}{manzanas rojas}
\openbox[labelalign=left,color=purple]{mi}{de Juan}
\openbox[rightend=false,rightarrow,color=orange!80!black]{od}{manzanas}
```

(7) 

(8) 

(9) 

(10) 

6.3 Floating labels: `\tagabove` and `\tagbelow`

```
\tagabove[<keys>]{<label>}{<text>}
\tagbelow[<keys>]{<label>}{<text>}
```

Example. Floating labels

```
\tagabove{or}{Juan compró manzanas}
\tagbelow{md}{las}
\tagbelow[tagx=0.5ex,tagy=2.2ex,color=blue]{md}{las}
```

(11) Juan compró^{OR} manzanas

(12) las

(13) 

6.4 Underline: \synul

```
\synul[<keys>]{<text>}
```

Example. Underline

```
\synul{compró}
\synul[color=red,underw=1.2pt]{compró}
```

(14) compró

(15) compró

7 Shortcut commands

Upper-span shortcuts are wrappers around \spanline: \SES, \ST, \SEC, \PVS, \PVC, and \OS.

Example. Upper shortcut commands

```
\SES{Juan} \PVS{compró manzanas}
\OS{\SES{Juan} \PVS{compró manzanas}}
```

(16) $\overset{\text{SES}}{\text{Juan}} \overset{\text{PVS}}{\text{compró manzanas}}$

(17) $\overset{\text{SES}}{\text{Juan}} \overset{\text{OS PVS}}{\text{compró manzanas}}$

Many lower shortcuts are dual commands. Without a star they produce a lower tag; with a star they produce a lower open box.

```
\OD{manzanas} % lower tag + underline
\OD*{manzanas} % lower open box
\MI{de Juan} % lower tag
\MI*{de Juan} % lower open box
```

(18) manzanas

(19) $\underset{\text{OD}}{\text{manzanas}}$

(20) de Juan

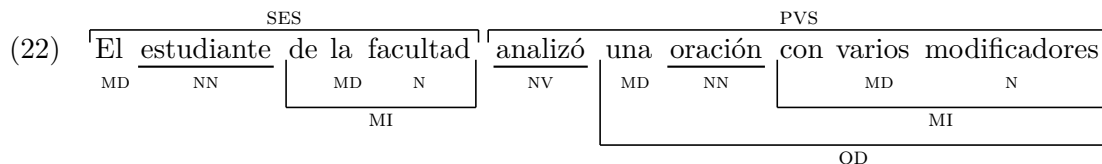
(21) $\underset{\text{MI}}{\text{de Juan}}$

Command	Output
SES, ST, SEC, PVS, PVC, OS	Upper spans.

OD, OI	Without star: lower label with underline. With star: lower open box.
CCL, COMP, MD, MI, CAg, CREG, CSim, CL, CM	Without star: lower tag. With star: lower open box.
PSO, PSNO, POO, PONO	Without star: lower tag. With star: lower open box.
ACT, ACL, ACM, ACF, ACC, ACCant, ACI, ACFin	Without star: lower tag. With star: lower open box.
MG, Ap, MF, IPCR, Incl	Without star: lower tag. With star: lower open box.
N, NN, NV, NP, NA, NAD	Simple lower labels; NN and NV include underline.
MGV, ApV, MFV, IPCRV, InclV	Lower labels with underline.

Example. Complex shortcut example

```
\SES{\MD{El} \NN{estudiante} \MI*{de \MD{la} \N{facultad}}}  
\PVS{\NV{analizó} \OD*{\MD{una} \NN{oración} \MI*{con \MD{varios} \N{modificadores}}}}
```



8 Manual multiline helpers

If a long span or open box must be split across visual lines, use the manual helpers. They draw separate local segments and avoid diagonal lines.

```
\spanfirst[<keys>]{<label>}{<text>} % left end, open to the right  
\spanmid [ <keys>]{<label>}{<text>} % continuation, no ends  
\spanlast [ <keys>]{<label>}{<text>} % open from the left, right end  
  
\boxfirst[<keys>]{<label>}{<text>}  
\boxmid [ <keys>]{<label>}{<text>}  
\boxlast [ <keys>]{<label>}{<text>}
```

Example. Manual upper split

```
\spanfirst{pvs}{propuso una consigna extensa}
\spanmid{pvs}{para analizar varios casos}
\spanlast{pvs}{en la clase práctica}
```

(23)
$$\overbrace{\text{propuso una consigna extensa}}^{\text{PVS}}$$

$$\overbrace{\text{para analizar varios casos}}^{\text{PVS}}$$

$$\overbrace{\text{en la clase práctica}}^{\text{PVS}}$$

Example. Manual lower split

```
\boxfirst{od}{una consigna extensa}
\boxmid{od}{con varios incisos}
\boxlast{od}{para resolver en clase}
```

(24)
$$\underbrace{\text{una consigna extensa}}_{\text{OD}}$$

$$\underbrace{\text{con varios incisos}}_{\text{OD}}$$

$$\underbrace{\text{para resolver en clase}}_{\text{OD}}$$

The upper shortcut wrappers are also available: `\SESfirst`, `\SESmid`, `\SESlast`, `\STfirst`, `\STmid`, `\STlast`, `\SECfirst`, `\SECmid`, `\SEClas`, `\PVSfirst`, `\PVSmid`, `\PVSlas`, `\PVCfirst`, `\PVCmid`, `\PVClas`, `\OSfirst`, `\OSmid`, and `\OSlast`.

9 Line wrapping and no-break variants

`\SMlinewrap` places its contents in a variable-width paragraph of maximum width `\linewidth`. It allows long examples to wrap naturally.

```
\SMlinewrap{El \MI{departamento de unos monoblocks deteriorados} quedaba abajo.}
```

`\openboxNB` and `\spanlineNB` wrap the annotated text in `\mbox`, preventing the marked text from breaking across lines.

```
\openboxNB{mi}{departamento de unos monoblocks deteriorados}
\spanlineNB{ses}{El departamento de unos monoblocks deteriorados}
```

(25)
$$\underbrace{\text{El departamento de unos monoblocks deteriorados}}_{\text{MI}} \text{ quedaba abajo.}$$

10 The `\breakspan` helper

`\breakspan` applies a right-opening style to all spans and boxes in its first argument and a left-opening style to all spans and boxes in its second argument.

```
\breakspan
{\SES{\MI{El departamento de unos monoblocks}}}
{\SES{\MI{quedaba en la planta baja}}}
```

(26)
$$\overbrace{\text{El departamento de unos monoblocks}}^{\text{SES}} \overleftarrow{\text{quedaba en la planta baja}}^{\text{SES}}$$

$$\text{MI} \qquad \text{MI}$$

11 Experimental automatic splitting

`\autospanline` and `\autoopenbox` require LuaLaTeX. They take plain text, compute a balanced line split, and internally emit the manual multiline helpers.

```
\autospanline[<width>]{<label>}{<plain text>}
\autoopenbox[<width>]{<label>}{<plain text>}
```

Example. Automatic split of an upper span

```
\autospanline[7cm]{pvs}{propuso una consigna extensa para analizar varios casos en la
clase práctica}
```

$$\overbrace{\text{propuso una consigna}}^{\text{PVS}}$$

$$\overbrace{\text{extensa para analizar}}^{\text{PVS}}$$

$$\overbrace{\text{varios casos en la clase práctica}}^{\text{PVS}}$$

Example. Automatic split of a lower box

```
\autoopenbox[7cm]{od}{una consigna extensa con varios incisos para resolver en la clase
práctica}
```

$$\underbrace{\text{una consigna extensa con varios}}_{\text{OD}}$$

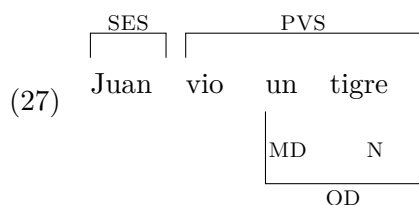
$$\underbrace{\text{incisos para resolver en la clase práctica}}_{\text{OD}}$$

These commands are experimental and do not parse nested `sintaxismark` markup.

12 Experimental grid mode

The `sintaxisgrid` environment reads a full nested structure, extracts tokens, computes line breaks, and draws the whole analysis as a single TikZ picture. It is the preferred mode for long examples in which classic spans or boxes would cross line breaks.

```
\begin{sintaxisgrid}[width=.92\linewidth,valign=base,yoffset=0pt]
\SA{
  \SES{Juan}
  \PVS{
    vio
    \OD*{
      \MD{un}
      \N{tigre}
    }
  }
}
\end{sintaxisgrid}
```



12.1 Grid environment options

Option	Meaning
<code>width=<dimension></code>	Available width for line breaking. Default: <code>\linewidth</code> .
<code>valign=base</code>	Baseline alignment; recommended inside <code>gb4e</code> .
<code>valign=t</code>	Top alignment.
<code>valign=c</code>	Center alignment.
<code>valign=b</code>	Bottom alignment.
<code>yoffset=<dimension></code>	Manual vertical adjustment. Default: <code>0pt</code> .

12.2 Grid interpretation of shortcut commands

Inside `sintaxisgrid`, commands are not executed in the classic sense. They are parsed structurally.

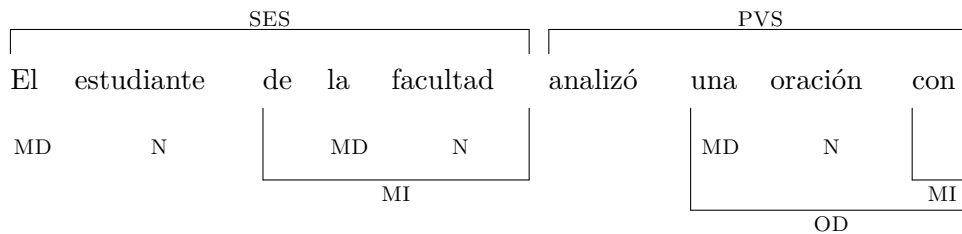
Command pattern	Grid interpretation
SES, ST, SEC, PVS, PVC, OS	Upper spans.
OD, OI	Lower underlined labels.
Any starred shortcut, e.g. OD*, MI*, COMP*	Lower open box.
Other unstarred shortcuts, e.g. MD, MI, N, NN, NV, ACL	Lower simple tags.

12.3 Grid spacing

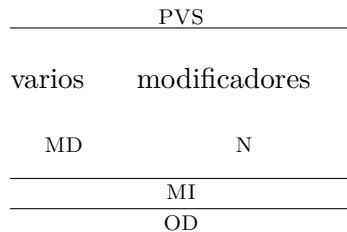
The grid renderer currently uses fixed internal constants tuned to resemble classic `\spanline` and `\openbox`: upper spans approximate `liney/labsep`; lower boxes approximate `boxd/boxsep`; lower tags approximate `tagy`; nested lower levels approximate `nestedgap` and the classic automatic depth behavior.

Example. Nested lower marks in grid mode

```
\begin{sintaxisgrid}[width=.92\linewidth,valign=base]
\SA{
  \SES{\MD{El}\N{estudiante}\MI*{de \MD{la}\N{facultad}}}}
  \PVS{analizó \OD*{\MD{una}\N{oración}\MI*{con \MD{varios}\N{modificadores}}}}
}
\end{sintaxisgrid}
```

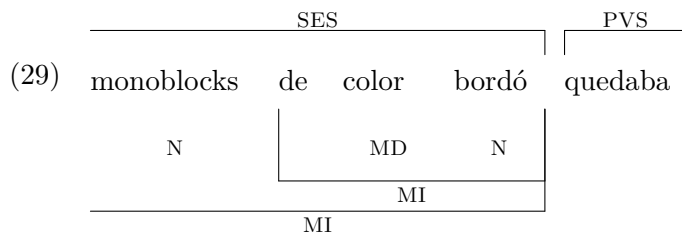
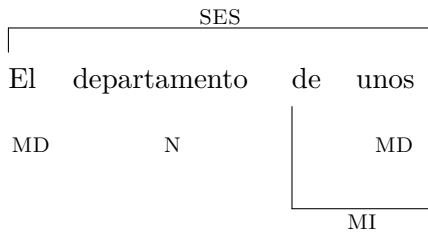


(28)



Example. Forcing a line break by narrowing the grid

```
\begin{sintaxisgrid}[width=8cm,valign=base]
\SA{
  \SES{\MD{El}\N{departamento}\MI*{de \MD{unos}\N{monoblocks}\MI*{de \MD{color}\N{bordó}}}}
  \PVS{quedaba \ACM{abajo}}
}
\end{sintaxisgrid}
```



The grid renderer emits its generated TikZ code without embedded newline character tokens, so normal use should not produce missing-character warnings.

13 Recommended workflow

1. Use classic mode for short examples and for precise manual control.
2. Use `\openboxNB`, `\spanlineNB`, or the manual split helpers when an annotated classical span must not cross a line break.
3. Use `\autospanline` or `\autoopenbox` only for plain text.
4. Use `sintaxisgrid` for long examples with nested syntactic markup.

14 Version notes

- **0.7.3 experimental:** adds formal package metadata, restores classic-mode compatibility with non-Lua engines, and removes spurious newline-character warnings from the grid renderer.
- **0.7.2 experimental:** monolithic file. `sintaxismark.sty` contains classic mode, automatic split helpers, and grid mode with embedded Lua.
- **0.7.1 experimental:** fixes body capture in `sintaxisgrid`.
- **0.7.0 experimental:** introduces `sintaxisgrid`.
- **0.6.23:** adds `\autospanline` and `\autoopenbox`.
- **0.6.22:** adds manual multiline helpers.

- **0.6.21:** improves nested `\tagbelow` depth handling.